

Q1 JEE Main 2020 - 6 September (Morning)

An insect is at the bottom of a hemispherical ditch of radius 1m. It crawls up the ditch but starts slipping after it is at height h from the bottom. If the coefficient of friction between the ground and the insect is 0.75, then h is ($g = 10ms^{-2}$)

- (A) 0.45m
- (B) 0.80m
- (C) 0.20m
- (D) 0.60m

Q2 JEE Main 2020 - 6 September (Evening)

A particle moving in the xy plane experiences a velocity dependent force $\vec{F} = k(v_y \hat{i} + v_x \hat{j})$, where v_x and v_y are the x and y components of its velocity $\vec{v}$. If $\vec{a}$ is the acceleration of the particle, then which of the following statements is true for the particle?

- (A) Quantity $\vec{v} \cdot \vec{a}$ is constant in time
- (B) Kinetic energy of particle is constant in time
- (C) Quantity $\vec{v} \times \vec{a}$ is constant in time
- (D) $\vec{F}$ arises due to a magnetic field

Q3 JEE Main 2020 - 7 January (Evening)

A mass of 10 Kg is suspended by a rope of length 4 m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals : (Take $g = 10ms^{-2}$ and the rope to be massless)

- (A) 100 N
- (B) 90 N
- (C) 75 N
- (D) 70 N

Q4 JEE Main 2020 - 7 January (Evening)

The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that

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$$|\vec{R}| = |\vec{P}|$$

. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is

Q5 JEE Main 2020 - 3 September (Morning)

A cricket ball of mass 0.15kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of 0.2m while launching the ball, the value of F (in N) is ($g = 10\text{ms}^{-2}$) _____.

Q6 JEE Main 2020 - 4 September (Evening)

A small ball of mass m is thrown upward with velocity u from the ground. The ball experiences a resistive force mkv^2 , where v is its speed. The maximum height attained by the ball is

- (A) $\frac{1}{2k} \tan^{-1} \frac{ku^2}{g}$
 (B) $\frac{1}{k} \ln \left(1 + \frac{ku^2}{2g} \right)$
 (C) $\frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$
 (D) $\frac{1}{k} \tan^{-1} \frac{ku^2}{2g}$

Q7 JEE Main 2020 - 8 January (Morning)

A particle of mass m is fixed to one end of light spring having force constant k and unstretched length l_0 . The other end is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is:

- (A) $\frac{l_0 m \omega^2}{k - m \omega^2}$
 (B) $\frac{l_0 m \omega^2}{k + m \omega^2}$
 (C) $\frac{l_0 m \omega^2}{k - m \omega}$
 (D) $\frac{l_0 m \omega^2}{k + m \omega}$

Laws Of Motion

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Answer Key

Q1 (C)

Q2 (C)

Q3 (A)

Q4 (90)

Q5 (150)

Q6 (C)

Q7 (A)

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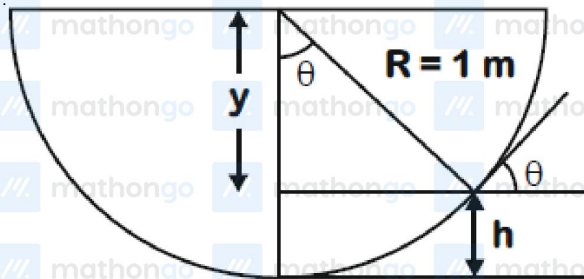
Laws Of Motion

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1



$$\mu = \tan \theta$$

$$\Rightarrow \frac{3}{4} = \tan \theta$$

$$\Rightarrow \theta = 37^\circ$$

$$\therefore h = R - R \cos \theta = 1 - 1 \times \frac{4}{5} = 0.2 \text{ m}$$

Q2

$$\frac{dv_x}{dt} = \frac{k}{m} v_y$$

$$\frac{dv_y}{dt} = \frac{k}{m} v_x$$

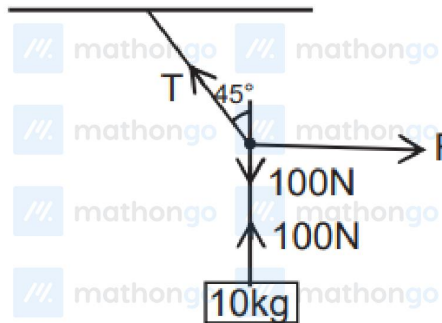
$$\Rightarrow \frac{dv_x}{dv_y} = \frac{v_y}{v_x} \Rightarrow v_x dx = v_y dy$$

$$\Rightarrow v_x^2 = v_y^2 + c$$

$$\vec{v} \times \vec{a} = (v_x^2 - v_y^2) \frac{k}{m}$$

$$\vec{v} \times \vec{a} = \frac{kC}{m}$$

Q3

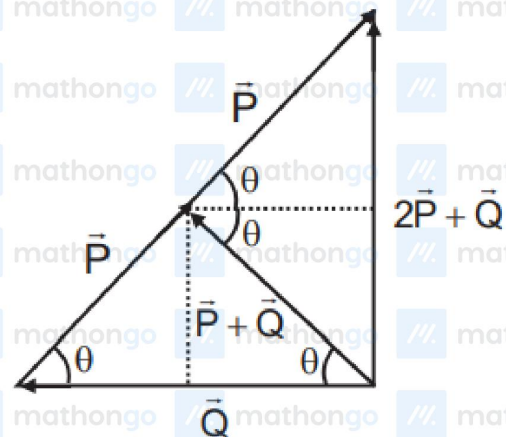


$$\frac{T}{\sqrt{2}} = 100$$

$$\frac{T}{\sqrt{2}} = F$$

$$F = 100 \text{ N}$$

Q4



So angle between $(2\vec{P} + \vec{Q})$ and $\vec{Q}$ is 90°

Q5

Initial velocity $V_0^2 = 2gh$

$$\Rightarrow v_0 = \sqrt{2 \times 10 \times 20} = 20 \text{ m/s}$$

Now work done by the machine.

$$F \cdot x = \Delta KE = \frac{1}{2} mv_0^2$$

Laws Of Motion

Hints & Solutions

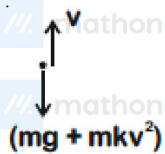
JEE Main 2020 Chapterwise

Physics

$$F \times \frac{2}{10} = \frac{1}{2} \times \frac{15}{100} \times 400$$

$$\Rightarrow F = 150$$

Q6



$$|a| = g + kv^2$$

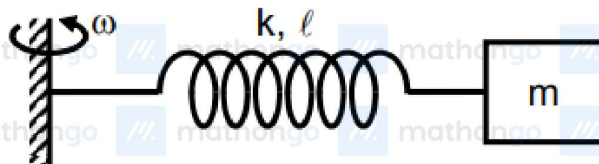
$$\Rightarrow -\frac{v dv}{dh} = g + kv^2$$

$$\Rightarrow \int_u^0 \frac{v dv}{g + kv^2} = \int_0^{H_{\max}} -dh$$

On solving

$$H_{\max} = \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$$

Q7



$$m\omega^2(\ell_0 + x) = kx$$

$$\left(\frac{\ell_0}{x} + 1 \right) = \frac{k}{m\omega^2}$$

$$x = \frac{\ell_0 m \omega^2}{k - m \omega^2}$$